# Taller 2 - Algebra Lineal

1. Describe the column space of these matrices: a point, a line, a plane, all of 3D space. Explain why.

   a) $$\boldsymbol{A}_1 = \begin{bmatrix} 2 & 2 \\ 1 & 1 \\ 5 & 6 \end{bmatrix}$$

   b) $$\boldsymbol{A}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

   c) $$\boldsymbol{A}_3 = \begin{bmatrix} 1 & 5 \\ 2 & 10 \\ 1 & 5 \end{bmatrix}$$

   d) $$\boldsymbol{A}_4 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

2. Compute $\boldsymbol{Ax}$ for the following matrix-vector pairs

   a) $$\boldsymbol{A} = \begin{bmatrix} 2 & 1 & 2 \\ 4 & 2 & 4 \\ 0 & 1 & 0 \end{bmatrix}, \quad \boldsymbol{x} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

   b) $$\boldsymbol{A} = \begin{bmatrix} -2 & 5 & 3 & 2 \\ -1 & -3 & -2 & 4 \end{bmatrix}, \quad \boldsymbol{x} = \begin{bmatrix} -3 \\ 2 \\ 0 \\ -6 \end{bmatrix}$$

   c) $$\boldsymbol{A} = \begin{bmatrix} -2 & 5 \\ -1 & -3 \\ 7 & 2 \\ -2 & 4 \end{bmatrix}, \quad \boldsymbol{x} = \begin{bmatrix} 10 \\ -10 \end{bmatrix}$$

   d) $$\boldsymbol{A} = \begin{bmatrix} -2 & 5 & 2 & -2 \\ -1 & -3 & 5 & 0 \\ 7 & 2 & 0 & -3 \\ -2 & 4 & -6 & 1 \end{bmatrix}, \quad \boldsymbol{x} = \begin{bmatrix} 2 \\ 4 \\ 6 \\ 8 \end{bmatrix}$$

3. Compute $\boldsymbol{AB}$ for the following matrices

   a) $$\boldsymbol{A} = \begin{bmatrix} 2 & 1 & 2 \\ 4 & 2 & 4 \\ 0 & 1 & 0 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

   b) $$\boldsymbol{A} = \begin{bmatrix} 2 \\ 4 \\ -3 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} -3 & 5 & 2 \end{bmatrix}$$

   c) $$\boldsymbol{A} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}$$

   d) $$\boldsymbol{A} = \begin{bmatrix} 2 & 4 \\ -2 & 1 \\ 3 & -3 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 5 & -3 & 2 & -7 \\ -1 & 4 & 0 & 5 \end{bmatrix}$$

4. Compute $\boldsymbol{ABC}$ for the following matrices

   a) $$\boldsymbol{A} = \begin{bmatrix} 2 & 1 & 2 \\ 4 & 2 & 4 \\ 0 & 1 & 0 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad \boldsymbol{C} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

b)

$$\boldsymbol{A} = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{C} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$$

c)

$$\boldsymbol{A} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \quad \boldsymbol{C} = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$$

d)

$$\boldsymbol{A} = \begin{bmatrix} 2 & 4 \\ -2 & 1 \\ 3 & -3 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 5 & -3 & 2 & -7 \\ -1 & 4 & 0 & 5 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

5. Factor these matrices into $\boldsymbol{A} = \boldsymbol{CR}$

a) $\boldsymbol{A} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

b) $\boldsymbol{A} = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & 1 & 3 & 5 \end{bmatrix}$

c) $\boldsymbol{A} = \begin{bmatrix} 2 & 1 & 3 \\ 6 & 3 & 9 \end{bmatrix}$

d) $\boldsymbol{A} = \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 2 & 2 & 0 \end{bmatrix}$